

Deadlock-Free Adaptive Routing and Bubble Flow Control for a 3D Torus

gem5 Garnet Lab 4 Project

Changxun Pan Liming Wang

Computer Architecture

August 2026

潘长浚

2024011323

王立明

2024011338

Problem and Scope

- A 3D Torus shortens paths with wrap links, but those links create cyclic channel dependencies.
- Topic 1: add 3D Mesh/Torus topologies and a deadlock-free minimal adaptive Torus router.
- Topic 2: add Critical Bubble Scheme (CBS) flow control to protect deterministic Torus DOR without reserving a VC.

Central idea

Keep a connected, acyclic progress path at the routing layer, or preserve one movable free slot at the flow-control layer.

3D Torus Implementation

- Parameterized Mesh3D.py and Torus3D.py.
- Evaluation instance: $4 \times 4 \times 4 = 64$ routers.
- Router ID: $zXY + yX + x$.
- 3D Mesh: 144 undirected neighbour pairs.
- 3D Torus: 192 pairs, adding 33.3% wrap connectivity.

ID	Policy
3	Torus minimal XYZ DOR
4	Torus adaptive + Escape VC
5	Mesh non-wrap XYZ

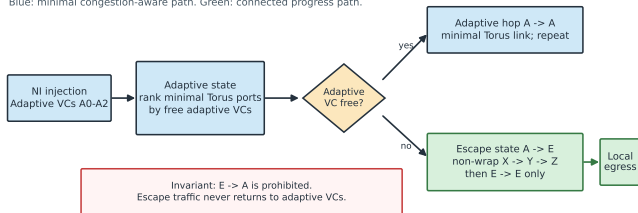
Controlled comparison

Torus DOR versus Torus adaptive keeps the topology, clocks, packet type, and total VC count fixed.

Topic 1: Adaptive Routing with Escape VCs

VC-class routing state machine

Blue: minimal congestion-aware path. Green: connected progress path.

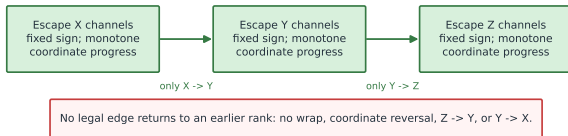


- Default: 3A/1E per vnet.
- A chooses a minimal Torus direction.
- E uses non-wrap XYZ.
- $A \rightarrow A$, $A \rightarrow E$, $E \rightarrow E$; never $E \rightarrow A$.

Why Escape VCs Are Deadlock-Free

Escape-channel dependency graph: a strict ordering

Every legal dependency advances dimension or monotone coordinate rank.



- No wrap link or direction reversal.
- Dimension order: X, Y, then Z.
- Escape CDG is acyclic.
- Escape cannot return to adaptive channels.

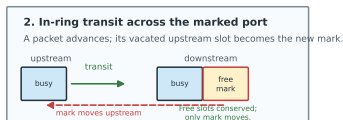
Result

A finite cyclic wait cannot remain after entering the Escape subnetwork.

Topic 2: Critical Bubble Scheme (CBS)

Critical Bubble Scheme: preserve one movable empty slot

A marked free VC is protected at ring entry and relocates after in-ring transit.



- DOR only, gated by `-enable-cbs`.
- Preserve one marked free buffer slot per directed ring.
- Ring entry needs two free VCs.
- In-ring transit moves the mark upstream.

Two Complementary Safety Mechanisms

Mechanism	Protected mode	Resource cost	Safety invariant
Escape VC	Adaptive Torus routing	Reserve $E \geq 1$ VC per vnet	Packets never return from Escape to adaptive VCs.
CBS	Torus DOR control traffic	No reserved VC	A marked free slot remains on every directed ring.

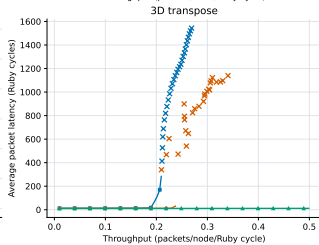
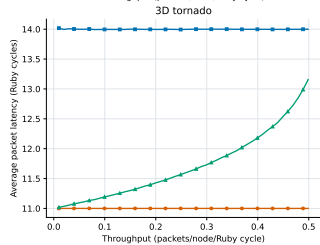
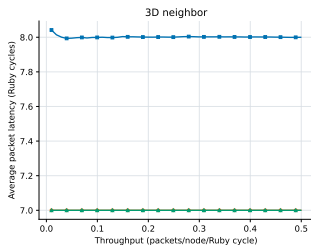
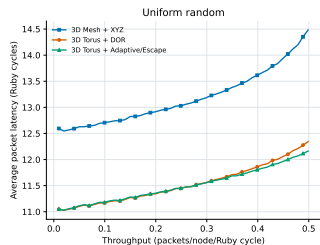
Different proof objects

Escape VC eliminates a channel-dependency cycle. CBS prevents a directed ring from becoming completely full.

Evaluation Methodology

- 64 routers, four VCs per vnet, single-flit vnet-0 control traffic.
- Main sweep: 10,000 Ruby cycles, four patterns: uniform random, neighbour, tornado, and transpose.
- Throughput is received packets/node/Ruby-cycle; source acceptance detects injection backpressure.
- Fixed-VC ablation: 4A/0E, 3A/1E, 2A/2E, 1A/3E.
- CBS sweep compares DOR baseline with DOR+CBS under the same settings.

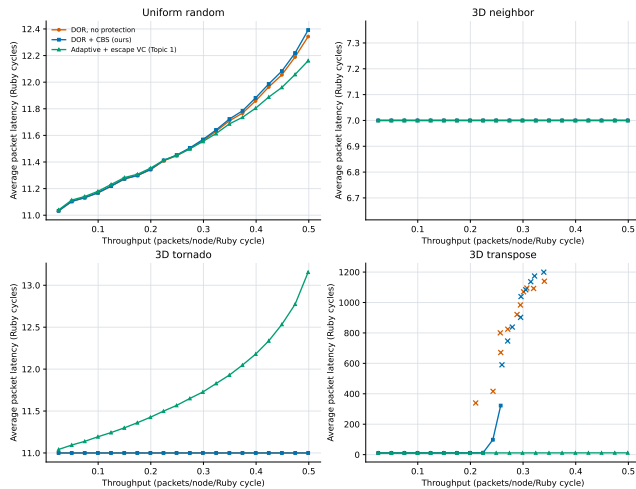
Topic 1 Results: Adaptive Routing



Mode	Stable TP	Hops
Mesh XYZ	0.210	3.763
Torus DOR	0.237	3.006
Adaptive	0.499	3.006

- Transpose traffic.
- Adaptive: 2.10x DOR throughput.
- Escape use stays low: 0.34% of transpose hops at full load.

Topic 2 Results: CBS Flow Control



- CBS keeps all four VCs available to DOR.
- On transpose, maximum stable throughput rises from 0.224 to 0.258 packets/node/Ruby-cycle.
- This is a 14.9% improvement.
- On other patterns, full-load latency changes by at most 0.049 cycles.

Division of Labor and Deliverables

潘长浔 2024011323

王立明 2024011338

Changxun Pan

- 3D Mesh/Torus topology and routing modes.
- Adaptive and Escape-VC design, experiments, and progress argument.
- Topic 1 figures and report sections.

Liming Wang

- CBS state, allocator rules, and statistics.
- Topic 2 experiments, figures, and correctness argument.
- Report integration and validation.

Shared

Evaluation methodology, code review, final report, and presentation.

Conclusion

- A 3D Torus implementation exposes useful path diversity but needs explicit deadlock control.
- Escape VCs provide a low-use, acyclic progress path for minimal adaptive routing.
- CBS provides an alternative for DOR: no reserved VC, but a conserved critical bubble per directed ring.
- Both mechanisms are implemented, measured, and accompanied by an invariant-based safety argument.

Questions